

REPÚBLICA DE MOÇAMBIQUE



Technical Circular

CT 170-001

SUBJECT: CONSTRUCTION OF VISUAL AND INSTRUMENT FLIGHT PROCEDURES

DATE: 21/11/2019

SECTION 1: GENERAL

1.0 Authority

This technical circular is issued under the authority of the Chairman of the Board and Chief Executive Officer of the Civil Aviation Institute of Mozambique (IACM), pursuant to Article 19 of Law 05/2016 of 14 June and the paragraph p) of Article 9 of decree 70/2017 of 30 December.

1.1 Purpose

This Technical Circular is issued to provide requirements of the approval of an organization for the design of Instrument Flight Procedures (IFP) and guidance to operators on the design, validation and publication of flight procedures.

1.2 Applicability

To air navigation service providers.

1.3 References

- | | |
|------------------------|---|
| a) ICAO Doc 8071 V1/V2 | <i>Testing ground-based Navaids / Satellite-based Navaids</i> |
| b) ICAO Doc 8168 V1/V2 | <i>Aircraft Operations – Flight Procedures</i> |
| c) ICAO Doc 8697 | <i>Aeronautical Chart manual</i> |
| d) ICAO Doc 9274 | <i>Manual on the Use of Collision Risk Model</i> |
| e) ICAO Doc 9365 | <i>Manual of All Weather Operation</i> |
| f) ICAO Doc 9906 V1-6 | <i>Flight Procedures Design</i> |
| g) ICAO Annex 4 | <i>Aeronautical Charts</i> |
| h) ICAO Annex 15 | <i>Aeronautical Information Services</i> |
| i) ICAO Doc 9905 | <i>Required Navigation Performance Authorization Required (RNP) Procedure Designer Manual</i> |
| j) ICAO Doc 10066 | <i>Aeronautical Information Management</i> |
| k) ICAO Doc 8697 | <i>Aeronautical Chart Manual</i> |
| l) ICAO Doc 10068 | <i>Manual on Development of a Regulatory Framework for Instrument Flight Procedure Design Service</i> |

1.4. Changes

This Technical Circular (TC) cancels the following Aeronautical Information Circulars (AICs):

- AIC 22/2011 – *Procedures Design*
- AIC 37/2012 – *Training Programme for Flight Procedures Designers*
- AIC 01/2014 – *Design, Validation and Testing of Flight Procedures*

1.5 Background

- a) MOZCAR 172 contains basic provisions on the design and validation of flight procedures.
- b) Uniform and systematic application of this guidance material shall ensure safety of the aircraft executing the procedures.
- c) Flight Procedures considered in this Technical Circular include conventional and RNAV departure, arrival and approach including non-precision and precision approaches.
- d) Additional guidance shall be sought from the ICAO reference documents listed under pt 1.3.

SECTION 2: Regulatory Process

2.1. General Guidelines

1. The Regulatory Process is based upon:

- a) IACM approval of procedure designers through evaluation of their training, experience, procedures and working practices;
- b) IACM regular periodic audit of procedure designers; and
- c) IACM evaluation and approval of completed IFP designs.

2. The IACM will only accept IFP design submissions from IACM Approved Procedure Designers (APDs).

3. IACM approval, once issued, will be non-transferable. All applications for recognition as a CAA APD will be judged solely on merit and compliance with the regulatory requirements. Where approval is withheld, the IACM will notify the applicant and provide a full explanation for the decision.

4. An approval certificate will be issued to successful applicants.

2.2. Designer Approval

An approved Design Organization shall comply with the requirements of this Circular at all times.

An approved Design Organization shall ensure the compliance of any third parties used whilst satisfying the requirements of this Circular.

The IACM considers that a combination of specialist knowledge and experience provides the basis for an effective and safe IFP designer which, when coupled to an effective management and documentation system, should ensure the output of a quality product that can be subjected to regulatory oversight.

2.3. Criteria for the Approval of IFP Designers

1. Procedure Designers seeking approval to design IFPs for use in the Mozambique airspace must provide evidence of the following:
 - a) Manual of procedures on PANS-OPS
 - b) Specialist PANS-OPS/IFP design training;
 - c) Practical Application of Theoretical Knowledge;
 - d) Aviation Experience; and
 - e) Quality Management System (QMS).
2. *Manual of procedures on PANS-OPS*: The IFP design organisation shall develop and maintain an operations manual. The operations manual shall serve to demonstrate how the service provider will comply with the requirements set out in this Circular – IFP Design.

The contents of the operations manual shall include but not limited to the following:

- (a) The information required of the IFP design organisation as mentioned in this Circular; and
- (b) A description of the IFP design office that shows the role, responsibilities and job functions of the IFP design office personnel who are responsible for ensuring the compliance of the organisation with the requirements in sub-paragraph (a).

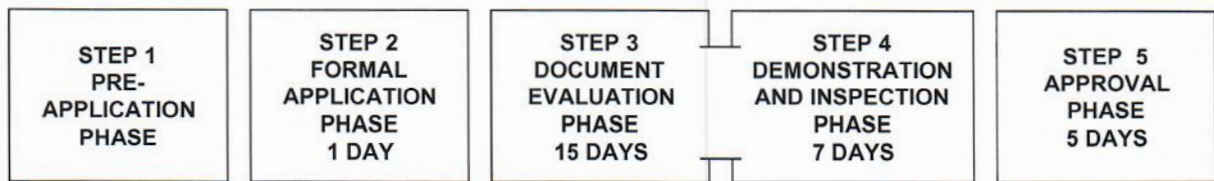
The IFP design organisation shall:

- (a) Keep the operations manual in a readily accessible form;
- (b) Ensure that the IFP designer has ready access to the operations manual; and
- (c) Amend the operations manual whenever necessary to keep its content up to date.

The IFP design organisation shall submit a copy of the most current operations manual to IACM.

3. *Specialist Training:* Proof of attendance and successful completion of a PANSOPS training course based upon ICAO Pans Ops Doc 8168. A typical PANS-OPS course is based on 4 to 8 weeks training, or equivalent part-time basis, given by an experienced lecturer, who is well grounded in procedure design and all aspects of PANS-OPS.
4. *Practical Application of Theoretical Knowledge:* The ability of an applicant to demonstrate practical application of theoretical knowledge is required. Applicants are expected to provide a proof of recent IFP design work; this should include details of specific designs that have been completed and over what period of time. Where possible, examples of the design process should be provided.
5. *Aviation Experience:* It is generally accepted that a high level of aviation experience is an important attribute for successful IFP design, ideally as aircrew, air traffic controller, Engineer or Aeronautical Information Officer.
6. *Quality Management System:* Applicants shall demonstrate that they have established and are able to maintain a documented quality system. This quality system shall be such that it enables the organisation to ensure that each design or any advice given with respect to any IFP issue conforms to international or national requirements and thus exercise the privileges as granted by their Approval. The quality system shall be described in a quality manual that includes control procedures for:
 - i) Management responsibility;
 - ii) A Quality System including:
 - Controlled documentation of the design process;
 - Record control system of design drawings and worksheets;
 - Record control system of input data including items such as:
 - Survey data and charting;
 - Record control system of regulatory documents and reference material;
 - Control of non-conforming design;
 - Records of personnel competence and qualifications;
 - Training of personnel;
 - Internal quality audits and corrective actions;
 - Subcontractor assessment audit and control; and,
 - Co-ordination throughout the process from design to notification with the sponsor for, or holder of, the design.

2.3.1 The Approval process may appear to be a complex undertaking, particularly to a “first-time” but are not complex. There are five steps in the provider approval process. Each step is shown for a detailed flow chart of the entire approval process:



2.3.2. In some cases, the guidance and suggested sequence of events and timeline in this TC may not be entirely appropriate. In such situations, the IACM and the operator should proceed in a manner that considers existing conditions and circumstances. The operator, however, should not expect to be certificated until the IACM is assured that the aviation law and its civil aviation regulations will be complied with in an appropriate and continuing manner.

2.4. Approval/Authorization of a Foreign

A need may arise for IACM to approve an IFP design organization that is located outside Mozambique. The procedure for approval of such foreign based ANSP-IFPD Organization should in principle cover the elements already mentioned in this Technical Circular for the approval.

2.5. Duration of approval

- (a) Design Organisation approval shall be valid for a maximum period of 5 years, unless it is previously suspended or revoked.
- (b) The holder of any approval that is suspended or revoked shall immediately surrender the approval certificate to the IACM.

2.6. Safety Inspections and Audits

- a. IACM shall conduct an initial certification audit and thereafter audits of IFPD - ANSP at intervals not exceeding 2 years (24 months) at the ANSP design office/facility.
 - (1) IFPD Organization audits shall be conducted in compliance with the IACM audit procedure;
 - (2) Foreign based IFPD Organization shall be responsible for bearing the cost of travel and Per Diem as per IACM policies in this regards;
 - (3) For each year IACM shall disseminate an audit program to ANSP detailing the Audit Schedule in order to allow ANSP to budget and plan for the audits.
- b. Failure to adhere to the applicable audit program and associated costs shall lead to the non-issuance or withdrawal of the certificate.

2.7. Resource Requirements

The IFP design organisation shall provide and maintain facilities for the design work on instrument flight procedures. This would include:

- a. Having available equipment appropriate for the design, design verification, flight validation, and maintenance of the types of instrument flight procedures;
- b. Access to relevant and current data including, but not limited to, aeronautical data, land

contour data, and obstacle data for the design, design verification, flight verification, and maintenance of the instrument flight procedures; and

- c. Ready access to copies of relevant documentation comprising technical standards, practices, and instructions, and any other documentation that may be necessary for the design, design verification, flight validation, and maintenance of the types of instrument flight procedure.

If an aeronautical database and aeronautical data is required for designing an instrument flight procedure, the IFP design organisation shall ensure the integrity of the database and the data. The data used shall be current, traceable, and meets the required level of verifiable accuracy for the design.

2.8. Documents and Records Control System

The IFP design organisation shall establish and put into effect, a system for controlling documents and records relating to the instrument flight procedures on which the designer carries on design work, including the policies and procedures for making, amending, preserving and disposing of those documents and records.

The IFP design organisation shall, at IACM's request, make the documents and records, or copies of them or extracts from them, available for inspection by IACM

2.9. Organizational Responsibilities for Flight Procedures

- a. The IFP design organisation is responsible for the recruitment and for the maintenance of competence of Flight Procedures designers and cartographers.
- b. The IFP design organisation is responsible for the planning, construction, testing, and validation of flight procedures.
- c. The IFP design organisation is responsible for the Quality Assurance in Flights Procedures design and publication.
- d. IACM is responsible for establishing the standards and for the approval of flight procedures.
- e. On receipt of a request for approval of a flight procedure for publication, IACM will evaluate the proposal including the supporting documentation and respond to the applicant.

Note: the matrix of organizational responsibilities for Standard Flight Procedures design and approval is shown in Attachment 1 to this Technical Circular.

2.10. Requirement for new or revised procedures

- a. Where an operational requirement exists for a new flight procedure, the IFP design organisation shall ensure that such procedure is designed in accordance with the prescribed standards and submitted to IACM for approval.
- b. The IFP design organisation may consult with IACM in advance or during the design process to clarify on regulatory requirements.

- c. Flight procedures published in the AIP should be revised:

CT 170-001 Construction of visual and instrument flight procedures

- I. When significant change to the obstacle environment occurs, requiring an amendment of procedural minimum altitudes;
 - II. When a published bearing, track or radial would fall into error by 1°, consequent on a change to magnetic variation or station declination;
 - III. To improve safety or operational efficiency, as identified by an interested party;
 - IV. To accommodate changes to aircraft category or characteristics;
 - V. To accommodate route connectivity on airspace organization change;
 - VI. Due to changes to the supporting navigation facility;
 - VII. To comply with amendments to applicable ICAO provisions and other international and national standards and recommended practices;
 - VIII. When a significant change occurs to aerodrome physical characteristics such as runways;
 - IX. When any other significant change occurs to aeronautical or topographical data.
- d. Each procedure should be re-assessed at least yearly and a revision proposed if necessary.

2.11. Non-Compliance

- a. Non-compliance with this regulation may require the IACM to restrict, suspend or revoke the IFPD Certificate.

SECTION 3. IFP Approval

3.1. Process and Criteria for the Submission of IFP Designs for Approval

- a) The design process is initiated by the ANSP's requirement for a new or change to an existing IFP.
- b) The Design Criteria for IFP design in the Mozambique airspace is based on the ICAO Doc 8168-OPS/611, Procedures for Air Navigation Services – Aircraft Operations Volume II, Construction of Visual and Instrument Flight Procedures (PANS-OPS Vol II)

3.2. Design Submission – Format and Content

3.2.1 IFP designs submitted for evaluation and approval by the regulator are to provide:

- a) A complete record of the design process including copies of all source data, information, calculations and drawings used in the project;
- b) A record of Quality Assurance and Quality Control;
- c) A statement of compliance with PANS-OPS from an IAPD;
- d) A report demonstrating how the original requirement has been satisfied;
- e) A narrative, which unambiguously describes the procedure in textual format and table showing all tracks in degrees True to 1/100th degree;
- f) A graphical representation which accurately reflects the content of the narrative provided;
- g) Relevant signed Validation reports; and
- h) A comprehensive design rationale in text format, including references to PANSOPS Volume II and National policy where a deviation from the standard criteria or policy has been employed.

3.2.2 Submissions

All completed submissions shall be sent to the Chairman of the Board and Chief Executive Officer.

3.3. PANSOPS Inspector Action

The inspector is expected to:

- a) Check the conformity of the IFP to DOC 8168 and other ICAO publications on the construction, design and quality assurance of procedures.
- b) Approve or decline giving reasons as to why.
- c) Communicate the verdict to the ANSP within three weeks from the date of application.
- d) Where the IACM is unable to approve an IFP design, consultation between the IACM and the ANSP will be encouraged to explore all possible solutions.

SECTION 4 Proficiency and training of procedures designer

4.1 Proficiency

In order to ensure that flight procedures submitted to IACM for approval and publication meet the required standards of quality assurance, the proficiency of the designers is specified as follows:

- a. Successful completion of an ICAO PANS-OPS course for the relevant flight procedure type;
- b. A minimum of five years aviation experience as a pilot, air traffic controller or equivalent experience;
- c. Completion of a minimum of two flight procedures designs under the supervision of a competent procedure designer;
- d. Flight procedures submitted for approval should be accompanied by details of competence of the designer(s).

4.2 Training

The training for IFP designers shall establish and implement training programmer to include an initial training and recurrent training at periodic intervals.

4.2.1 Categories of personnel to be trained

The following categories of personnel are covered by this circular:

- a) Flight procedure designers
- b) Cartographers
- c) Flight Procedures inspectors

4.2.2 Approved training organizations

- a. A number of organizations provide training in flight procedure design. An IFP design organisation shall request the IACM to approve training organizations intended for the provision of training.
- b. The IFP design organisation shall bear the costs of approval.

4.2.3 Requirements for training

- a. Basic flight procedure design training

Basic flight procedure design shall be provided to all personnel to be involved in the design, development, drawing and flight inspection. The training should provide participants with basic skills to develop non-precision approach and departure procedures using the ICAO PANS-OPS Doc 8168 criteria. The training will form a foundation for progression in the areas of specialization for each individual.

b. Specialized training

Flight procedure design is a wide discipline and it is not practicable for an individual to cover all aspects of the subject. The ANSP needs to identify individual talents and provide specialized training as follows:

- I. Precision approach procedures
- II. RNAV and RNP procedures
- III. Quality control in flight procedure design
- IV. Flight procedures inspection

c. On-the-Job Training

- I. The IFP design organisation ANSP shall ensure that performance based OJT programmes involving designing, developing, drawing and flight inspection of procedures are conducted before any individual can be authorized to carry out specified duties.
- II. The ANSP shall engage qualified OJT instructors to carry out specialized training in the field.

d. Refresher training

- I. The technology employed in developing procedures as well as the airborne and space based system used in navigation are in a constant state of evolution. To cope with changing technology and operating procedures, it is necessary that the IFP design organisation provide personnel with courses of instruction designed to bring knowledge and skills up to date.
- II. Refresher courses may range from in-house to full-fledged training in specialized training organizations.

e. Recurrent training

- I. Considering the level of activities in an organization, it might not be cost effective to establish a fully-fledged flight procedure design unit. The personnel will only be called when a need to develop new procedures or review existing ones is identified.

- II. Recurrent training will be provided to ensure that the procedure designers remain current and that there is no need to retrain personnel following a long period of inactivity.

4.2.4 Training records

- a. The IFP design organisation shall ensure that training records, including OJT are properly kept for inspection by the IACM as may be required.
- b. The training records shall include certificates, OJT tasks performed and any other documents related to training and approval of jobs performed.

4.2.5 Requirement for approved curriculum

- a. The IFP design organisation shall develop training curricula for all types of training as required by this circular.
- b. The curricula shall be approved by the IACM.

4.2.5.1 Standards and criteria

- a. The PANS OPS Criteria contained in ICAO Doc 8168 shall be applied in developing instrument approach procedures and submitted to the IACM for approval.
- b. The design and format for IAP charts shall be in accordance with ICAO Annex 4 and ICAO Doc 8697.

4.2.5.2. Airspace organization

- a. Instruments flight paths shall be contained within controlled airspace (class A, B, C, D or E), where established.
- b. Where instrument flight paths are contained within controlled airspace which lies above uncontrolled airspace, the minimum procedural altitude should be at least 500 ft above the base of the controlled airspace.
- c. Any proposal to establish a terminal instrument flight path in uncontrolled airspace will require a safety assessment including consideration of types and density of air traffic, risk analysis and acceptable mitigation (ref IACM - CT 100-01).

SECTION 5 Design Procedures

5.1 Data acquisition

The information and data required for procedure design shall consist of:

- a. Airport, NAVAID, obstacle, terrain coordinates and elevation data based on verified surveys and compliant with ICAO Annex 11, 14 and 15 requirements;
- b. Airspace requirements;
- c. Airport infrastructure: runway classification, lighting, communications, runway markings and local altimeter setting;
- d. Environmental considerations;
- e. Any other potential issue associated with the procedure.

5.2 Review of design and approval

Each new or revised procedure shall be verified by a qualified procedure designer other than the one who designed the procedure, to ensure compliance with applicable criteria before submission to the IACM for approval. The credentials and the comments of the procedure designer reviewing the procedure must be attached to the documentation submitted for approval.

5.3 Documentation to be submitted

The procedure designer shall submit:

- a) The documentation required for publication in the AIP in accordance with ICAO Annexes 4 and 15
- b) The documentation required to maintain transparency concerning the details and assumptions used by the procedure designer, which should include supporting information and data used in the design, such as:
 - I. Controlling obstacle for each segment of the procedure
 - II. Effect of environmental considerations on the design of the procedure
 - III. Infrastructure assessment
 - IV. Airspace constraints
 - V. The results of the periodic reviews and, for modifications or amendments to existing procedures, the reasons for any change
 - VI. For any deviation from existing standards, the reasons for such a deviation, details of the safety assessment and of the mitigations applied to assure continued safe operation
 - VII. The results of the final verification for accuracy and completeness.
- c) Additional documentation required to facilitate ground and flight validation of the procedure and the results of ground and flight validation.

5.4 Retention of documentation

The entity responsible for developing flight procedures shall retain during the entire lifetime of the procedure and 24 months after its withdrawal all procedure design and review documentation so as to allow any data irregularities or errors found during the production, maintenance or operational use of the procedure to be corrected, and to support any incident/accident investigation.

5.5 Obstacle clearance altitude / height OCA / OCH

The entity responsible for developing flight procedures shall publish obstacle clearance altitude OCA and obstacle clearance height OCH for an aerodrome as described in ICAO Doc 8168 vol II.

5.6 Minimum Descent Altitude MDA and Minimum Descent Height MDH

The entity responsible for developing flight procedures shall establish and publish minimum descent altitude MDA and Minimum descent height MDH for an aerodrome as described in ICAO Annex 4 and Doc 8168 vol II. This applies also when the State has established operating minima for the aerodrome. (ICAO Doc 9365 chap 6).

5.7 Decision Altitude DA and Decision Height DH

The entity responsible for developing flight procedures shall establish and publish Decision Altitude DA and Decision Height DH for an aerodrome as described in ICAO Annex 4 and Doc 8168 vol II. This applies also when the State has established operating minima for the aerodrome. (ICAO Doc 9365 chap 6).

5.8 Validation of flight procedures

Validation is the necessary final quality assurance step in the procedures design process prior to publication. The purpose of validation is the verification of all obstacles and navigation data, and assessment of applicability of the procedure. Ground validation shall always be undertaken. When the IACM can verify, by ground validation, the accuracy and completeness of all obstacles and navigation data, then the IACM may decide to dispense with the flight validation requirement.

5.9 Ground validation

Ground validation is a review of the entire instrument flight procedure by a person trained in procedure design and with appropriate knowledge of flight validation issues. It is meant to catch errors in criteria and documentation, and evaluate on the ground, to the extent possible, those elements that will be evaluated in flight. Issues identified in the ground validation should be addressed prior to any flight validation. The ground validation will also determine if flight validation is needed for

modifications and amendments to previously published procedures.

5.10 Flight validation

- a) Flight validation of instrument flight procedures shall be carried out as part of the initial certification and shall be included as part of the periodic quality insurance program. It shall be performed by a flight crew approved by the IACM. The objectives of flight validation are:
 - I. Provide assurance that adequate obstacle clearance has been provided;
 - II. Verify that the navigation data used are correct;
 - III. Verify that all required infrastructures, such as runway marking, lighting, communication and navigational aids are in place and operative;
 - IV. Verify that the procedure can be safely flown;
 - V. Evaluate the charting, required infrastructures, visibility and other operational factors.
- b) A briefing to the flight crew shall be performed before the flight test by the designer of the procedure or by a qualified procedure designer having reviewed the procedure.
- c) To perform flight validation of procedures, pilots must carry Commercial Pilot License and a valid Instrument Rating with appropriate experience and be approved by the IACM.

5.11 Flight inspection

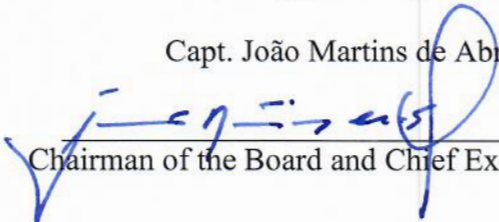
Flight inspection of instrument flight procedures shall not be confused with flight validation, and is required to assure that the appropriate radio navigation aids adequately support the procedure. This is carried out as a part of a formal flight inspection program and is performed by a qualified and certified flight inspector using an appropriately equipped and certified aircraft.

5.12 Periodical review of flight procedures

The entity responsible for flight procedures design shall establish, submit to IACM and conduct a program of periodical review of the procedures to ensure that they continue to comply with changing criteria and meet the user's requirements.

Approved by:

Capt. João Martins de Abreu


Chairman of the Board and Chief Executive Officer

CT 170-001 Construction of visual and instrument flight procedures

